

Name of Student: Vyankatesh Suresh Rohokale	PRN: 16070122518
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 16	
TITLE : User-Defined Exceptions	

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

Questions:

1. Create a Voting System that throws a custom exception if age is below 18
2. Create a Driving License System that throws a custom exception if the user's age is below 18. If the age is valid , display that the user is eligible for a driving license

Programme

1:

```

J Demo.java
1  import java.util.*;
2
3  class AgeException extends Exception {
4
5      AgeException(String message) {
6          super(message);
7      }
8  }
9  class Voting {
10     void checkAge(int age) throws AgeException {
11
12         if (age < 18) {
13             throw new AgeException("Age is below 18. Not eligible for voting.");
14         }
15         System.out.println("You are eligible for voting.");
16     }
17 }
18 public class Demo {
19     public static void main(String[] args) {
20         Scanner sc = new Scanner(System.in);
21
22         System.out.print("Enter your age: ");
23         int age = sc.nextInt();
24

```

```

24
25         try {
26             Voting v = new Voting();
27             v.checkAge(age);
28         }
29         catch (AgeException e) {
30             System.out.println("Error: " + e.getMessage());
31         }
32
33         sc.close();
34     }
35 }

```

2:

```

J Demo.java
1  import java.util.*;
2  class LicenseException extends Exception {
3      LicenseException(String message) {
4          super(message);
5      }
6  }
7  class DrivingLicense {
8      void checkEligibility(int age) throws LicenseException {
9          if (age < 18) {
10             throw new LicenseException(
11                 "Age is below 18. Not eligible for driving license."
12             );
13         }
14         System.out.println("User is eligible for a driving license.");
15     }
16 }
17
18 public class Demo {
19     public static void main(String[] args) {
20         Scanner sc = new Scanner(System.in);
21         System.out.print("Enter your age: ");
22         int age = sc.nextInt();
23         try {
24             DrivingLicense d = new DrivingLicense();
25             d.checkEligibility(age);
26         }
27         catch (LicenseException e) {
28             System.out.println("Error: " + e.getMessage());
29         }
30
31         sc.close();
32     }
33 }

```

Output

1:

```
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter your age: 18
You are eligible for voting.
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter your age: 12
Error: Age is below 18. Not eligible for voting.
piyushrohokale@Piyushs-MacBook-Air DSA assignments %
```

2:

```
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter your age: 18
User is eligible for a driving license.
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
1Enter your age: 11
User is eligible for a driving license.
piyushrohokale@Piyushs-MacBook-Air DSA assignments %
```

Github :

Java_assignments / assignment16 /

VyankateshRohokale Create DrivingLicenseSystem.java 1c8bdf0 · now History

Name	Last commit message	Last commit date
..		
.DS_Store	a14	26 minutes ago
1.png	a14	26 minutes ago
11.png	a14	26 minutes ago
2.png	a14	26 minutes ago
22.png	a14	26 minutes ago
DrivingLicenseSystem.java	Create DrivingLicenseSystem.java	now
VotingSystem.java	Create VotingSystem.java	now

Conclusion:

The programs successfully demonstrate user-defined exceptions in Java. Custom exception classes are created by extending the Exception class and are thrown when the user's age is below 18. The exceptions are then handled using try-catch , allowing application-specific error messages to be displayed.